

## SEMESTER END EXAMINATIONS - APRIL 2023

|             |                                                                 |            |                |
|-------------|-----------------------------------------------------------------|------------|----------------|
| Program     | : <b>B.E :- Common to CSE / ISE / AI &amp; ML / AI &amp; DS</b> | Semester   | : <b>III</b>   |
| Course Name | : <b>Linear Algebra and Laplace Transforms</b>                  | Max. Marks | : <b>100</b>   |
| Course Code | : <b>CS/IS/AI/AD31</b>                                          | Duration   | : <b>3 Hrs</b> |

### Instructions to the Candidates:

- Answer one full question from each unit.

### UNIT - I

- Find (i)  $L\left\{\left(\sqrt{t}-\frac{1}{\sqrt{t}}\right)^3\right\}$  (ii)  $L\left\{\int_0^t e^{-t} \cos t \, dt\right\}$ . CO1 (06)
  - Find (i)  $L\left\{2^t + \frac{\cos 2t - \cos 3t}{t} + t \sin t\right\}$ . CO1 (07)
  - Find the Laplace transform of the triangular wave function of period  $2a$ , CO1 (07)  
given by  $f(t) = \begin{cases} t & \text{for } 0 \leq t \leq a \\ 2a - t & \text{for } a \leq t \leq 2a \end{cases}$ .
- Find  $L\{\sin \sqrt{t}\}$ . CO1 (06)
  - If  $L\{f(t)\} = F(s)$  then prove that  $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$ , where  $n$  is a CO1 (07)  
positive integer.
  - Evaluate: CO1 (07)  
(i)  $\int_0^\infty \frac{e^{-3t} - e^{-2t}}{t} dt$  (ii) Find  $L\{e^{-3t} t^8\}$ .

### UNIT - II

- Find (i)  $L^{-1}\left\{\frac{s^3 + 6s^2 + 12s + 8}{s^6}\right\}$  (ii)  $L^{-1}\left\{\frac{s e^{-s/2} + \pi e^{-s}}{s^2 + \pi^2}\right\}$ . CO2 (06)
  - Find  $L^{-1}\left\{\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}\right\}$  using convolution theorem. CO2 (07)
  - The coordinates  $(x, y)$  of a particle moving along a plane curve at any CO2 (07)  
point  $t$ , are given by  $y' + 2x = \sin 2t$ ,  $x' - 2y = \cos 2t$ ,  $t > 0$ . Given  
 $x(0) = 1$  and  $y(0) = 0$  show that using Laplace Transforms the particle  
moves along the curve  $4x^2 + 4xy + 5y^2 = 4$ .

4. a) Find  $L^{-1}\left(\log\left(\frac{s^2+1}{s(s+1)}\right)\right)$ . CO2 (06)
- b) Express the function  $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$  in terms of the unit step function and hence find its Laplace transform. CO2 (07)
- c) Solve  $y''' + 2y'' - y' - 2y = 0$ , given that  $y(0) = 0$ ,  $y'(0) = 0$ ,  $y''(0) = 6$  by applying Laplace transforms. CO2 (07)

**UNIT - III**

5. a) Determine the matrix that defines a reflection in the y-axis followed by rotation through an angle  $\pi/6$  and then by the dilation of factor  $3/2$ . CO3 (06)  
Find the image of the point  $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$  under this transformation.
- b) Prove that the transformation  $T: R^2 \rightarrow R^3$  is defined by  $T(x, y) = (3x + y, 2y, x - y)$  is linear and hence find the images of  $(1, 2)$  and  $(2, -5)$  under this transformation. CO3 (07)
- c) Find the transition matrix  $P$  from the basis  $B$  to the standard basis  $B'$  of  $R^2$ ,  $B = \{(2, 3), (1, 2)\}$  and  $B' = \{(1, 0), (0, 1)\}$ . Use this matrix to find the coordinate vector  $u$  relative to  $B'$ , where  $u_B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ . CO3 (07)
6. a) Verify that the vector  $(4, -4, 6)$  is a linear combination of the vectors  $(1, 2, -3)$ ,  $(2, -4, 6)$  and  $(-1, 2, -3)$ . CO3 (06)
- b) Show that a linear operator  $T: R^2 \rightarrow R^2$  defined by  $T(X) = AX$  rotates a vector  $x$  through an angle  $\theta$  about the origin if  $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ . CO3 (07)
- c) Find the range and kernel of a linear transformation  $T: R^2 \rightarrow R^3$  is given by  $T(x, y) = (3x, x - y, y)$  and hence verify Rank - nullity theorem. CO3 (07)

**UNIT- IV**

7. a) Find the dimension and basis for four fundamental subspaces of CO4 (10)

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

- b) Find the least-squares solution of  $Ax = b$ , where CO4 (10)

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, b = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

8. a) Find the complete solution of  $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix}$ . CO4 (10)

b) Obtain the QR factorization of  $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ . CO4 (10)

**UNIT - V**

9. a) Diagonalize the matrix  $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$  and hence find  $A^4$ . CO5 (10)

b) (i) Determine whether the matrix  $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$  is CO5 (10)

Hermitian.

(ii) Verify the matrix  $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$  is positive definite?

10. a) Orthogonally diagonalize the matrix  $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ . CO5 (10)

b) Find a singular value decomposition of  $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$ . CO5 (10)

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